Lecture 13

A particle in space: orbital angular momentum

Objectives. *Orbital angular momentum acts on wavefunctions in three dimensions. We decompose their space into finite-dimensional multiplets and prove that only integer spins occur. Harmonic polynomials give spherical harmonics, and central potentials reduce to a radial rule. The oscillator provides two counts of the same level.*

- Count the integer-spin multiplets in an oscillator level.
- Construct spherical harmonics from harmonic polynomials.
- Derive the radial rule away from the origin for smooth radial functions.

13.1 Three dimensions

A point of space is $x = (x_1, x_2, x_3)$, with $r = |x|$. The space $L^2(\mathbb{R}^3)$ consists of the classes of square-integrable functions on $\mathbb{R}^3$, with $\langle \varphi, \psi \rangle = \int \overline{\varphi(x)}\psi(x)\,\mathrm{d}^3x$. As in Definition 4.1, $\mathcal{S}(\mathbb{R}^3)$ is the set of infinitely differentiable functions ψ such that every product of a monomial $x_1^{a_1}x_2^{a_2}x_3^{a_3}$ with a partial derivative of ψ is bounded. It is dense in $L^2(\mathbb{R}^3)$ by the standing fact S4. The Fourier transform is given by

$$\tilde{\psi}(p) = (2\pi\hbar)^{-3/2} \int e^{-\mathrm{i}p\cdot x/\hbar}\psi(x)\,\mathrm{d}^3x.$$

By the standing fact S1 it extends to a unitary operator F on $L^2(\mathbb{R}^3)$.

Definition 13.1 (Position, momentum and free energy in space)**.** On $L^2(\mathbb{R}^3)$, for $j = 1, 2, 3$, the position Q_j multiplies by x_j on $D(Q_j) = \{\psi : x_j\psi \in L^2(\mathbb{R}^3)\}$, the momentum is $P_j = F^{-1}M_{p_j}F$, and the free energy of a particle of mass m is $H_0 = F^{-1}M_{|p|^2/2m}F$, each on the vectors whose image under F the multiplier keeps square integrable.

Proposition 13.2 (Three dimensions)**.** *The operators Q_j, P_j and H_0 are self-adjoint, and H_0 has no eigenvalue. Every $\psi \in \mathcal{S}(\mathbb{R}^3)$ lies in their domains, with*

$$Q_j\psi = x_j\psi, \qquad P_j\psi = -\mathrm{i}\hbar\,\partial_j\psi, \qquad H_0\psi = -\frac{\hbar^2}{2m}\,\Delta\psi,$$

where $\Delta = \partial_1^2 + \partial_2^2 + \partial_3^2$. These operators map $\mathcal{S}(\mathbb{R}^3)$ into itself, and on it $Q_jP_k\psi - P_kQ_j\psi = \mathrm{i}\hbar\delta_{jk}\psi$, while the Q_j commute with each other and so do the P_j.

Proof. The proofs of Lemmas 0.7 and 0.8 hold word for word on $\mathbb{R}^3$, with the cube $[-n, n]^3$ in place of $[-n, n]$ and S4 for $\mathbb{R}^3$. So multiplication by a real continuous function on $\mathbb{R}^3$

is self-adjoint, and so are Q_j, P_j and H_0. Integration by parts in the variable x_j, as in Proposition 4.3, gives the transform $(\mathrm{i}p_j/\hbar)\tilde{\psi}$ of $\partial_j\psi$, hence the three actions. The product rule maps $\mathcal{S}(\mathbb{R}^3)$ into itself and gives the relations, as in Proposition 4.7. Partial derivatives commute, so the Q_j commute with each other and so do the P_j. Finally, if $H_0\psi = \lambda\psi$, then $(|p|^2/2m - \lambda)\tilde{\psi} = 0$. As in the last step of the proof of Lemma 0.7, $\tilde{\psi}$ vanishes on every closed cube with rational corners that misses the sphere $|p|^2 = 2m\lambda$. These cubes cover $\mathbb{R}^3$ except the sphere. A sphere has zero volume, so, like a point on the line, it does not change a class, and $\tilde{\psi} = 0$ and $\psi = 0$. ∎

A *rotation* is a real 3×3 matrix R with $R^{\mathrm{T}}R = \mathbb{1}$ and $\det R = 1$. It acts on wave functions by

$$\big(U(R)\psi\big)(x) = \psi(R^{-1}x).$$

The change of variables $y = R^{-1}x$ has Jacobian 1, so $U(R)$ is unitary, and $U(R)U(R') = U(RR')$. It maps $\mathcal{S}(\mathbb{R}^3)$ onto itself. We write $R_3(\alpha)$ for the rotation by the angle α about the x_3-axis, $R_3(\alpha)x = (x_1\cos\alpha - x_2\sin\alpha,\ x_1\sin\alpha + x_2\cos\alpha,\ x_3)$.

Self-check 13.1

1. Let R be a rotation and $\psi \in \mathcal{S}(\mathbb{R}^3)$. Show that $\Delta(U(R)\psi) = U(R)\Delta\psi$, so $U(R)$ commutes with H_0 on $\mathcal{S}(\mathbb{R}^3)$.

13.2 Orbital angular momentum

Definition 13.3 (Orbital angular momentum)**.** On $\mathcal{S}(\mathbb{R}^3)$, the *orbital angular momentum* $\mathbf{L} = Q \times P$ has the components

$$L_1 = Q_2P_3 - Q_3P_2, \qquad L_2 = Q_3P_1 - Q_1P_3, \qquad L_3 = Q_1P_2 - Q_2P_1.$$

We write $L_\pm = L_1 \pm \mathrm{i}L_2$ and $L^2 = L_1^2 + L_2^2 + L_3^2$.

By Proposition 13.2, the components act by differential rules, for example $L_3\psi = -\mathrm{i}\hbar(x_1\partial_2\psi - x_2\partial_1\psi)$. We use the same rules on every smooth function on an open set.

Proposition 13.4 (Orbital angular momentum)**.**

1. *The L_j map $\mathcal{S}(\mathbb{R}^3)$ into itself, and $\langle\chi, L_j\psi\rangle = \langle L_j\chi, \psi\rangle$ for all $\chi, \psi \in \mathcal{S}(\mathbb{R}^3)$.*
2. *On $\mathcal{S}(\mathbb{R}^3)$,*
$$[L_1, L_2] = \mathrm{i}\hbar L_3, \qquad [L_2, L_3] = \mathrm{i}\hbar L_1, \qquad [L_3, L_1] = \mathrm{i}\hbar L_2.$$
3. *Let $f(x) = G(|x|^2)$ with G smooth on an open interval. Then $L_jf = 0$ where f is defined, and $L_j(f\psi) = fL_j\psi$ there for every smooth ψ.*
4. *For $\psi \in \mathcal{S}(\mathbb{R}^3)$ and every x, $\frac{\mathrm{d}}{\mathrm{d}\alpha}\big(U(R_3(\alpha))\psi\big)(x)\big|_{\alpha=0} = -\frac{\mathrm{i}}{\hbar}(L_3\psi)(x)$.*

Proof.

1. By Proposition 13.2, the Q_k and P_l map $\mathcal{S}(\mathbb{R}^3)$ into itself, and they are symmetric there, since they are self-adjoint. For $k \neq l$, $Q_kP_l = P_lQ_k$ on $\mathcal{S}(\mathbb{R}^3)$, so $\langle\chi, Q_kP_l\psi\rangle = \langle P_lQ_k\chi, \psi\rangle = \langle Q_kP_l\chi, \psi\rangle$. Each L_j is a difference of two such products.
2. Expand $L_1L_2 - L_2L_1$ into four commutators. Since Q_k commutes with P_l for $k \neq l$, two of them vanish, and the other two are
$$[Q_2P_3, Q_3P_1] = Q_2\,[P_3, Q_3]\,P_1 = -\mathrm{i}\hbar Q_2P_1, \qquad [Q_3P_2, Q_1P_3] = Q_1\,[Q_3, P_3]\,P_2 = \mathrm{i}\hbar Q_1P_2.$$

So $[L_1, L_2] = \mathrm{i}\hbar(Q_1P_2 - Q_2P_1) = \mathrm{i}\hbar L_3$. The cyclic permutation $1 \to 2 \to 3 \to 1$ carries Definition 13.3 into itself, so it gives the other two relations.

3. By the chain rule, $x_k\partial_l f - x_l\partial_k f = 2x_kx_lG' - 2x_lx_kG' = 0$. The second claim is the product rule.
4. With $R_3(\alpha)^{-1}x = (x_1\cos\alpha + x_2\sin\alpha,\ -x_1\sin\alpha + x_2\cos\alpha,\ x_3)$, the chain rule gives the derivative $x_2\partial_1\psi - x_1\partial_2\psi$, which is $-(\mathrm{i}/\hbar)L_3\psi$. ∎

So L_3 generates the rotations about the x_3-axis, and the L_j satisfy the relations of an angular momentum (*Quantum Mechanics I*, Def. 7.1). But they act on $\mathcal{S}(\mathbb{R}^3)$, which is not finite-dimensional. The complex coordinate

$$w = x_1 + \mathrm{i}x_2$$

makes L_3 diagonal. Since $(x_1\partial_2 - x_2\partial_1)w = \mathrm{i}w$ and $(x_1\partial_2 - x_2\partial_1)\overline{w} = -\mathrm{i}\overline{w}$, the product rule gives

$$L_3\big(w^k\overline{w}^{k'}x_3^c\big) = \hbar(k-k')\,w^k\overline{w}^{k'}x_3^c \tag{13.1}$$

for all integers $k, k', c \geq 0$. In the same way $L_+ = \hbar\big(x_3(\partial_1 + \mathrm{i}\partial_2) - w\partial_3\big)$, and since $(\partial_1 + \mathrm{i}\partial_2)w = 0$, $L_+w^l = 0$ for every l. The rotation $R_3(\alpha)^{-1}$ multiplies w by $e^{-\mathrm{i}\alpha}$ and $\overline{w}$ by $e^{\mathrm{i}\alpha}$, so for every smooth function f of $|x|^2$,

$$U(R_3(\alpha))\big(w^k\overline{w}^{k'}x_3^c f\big) = e^{-\mathrm{i}\alpha(k-k')}\,w^k\overline{w}^{k'}x_3^c f. \tag{13.2}$$

Self-check 13.2

1. Find L_3 and L_+ on w, $\overline{w}$ and x_3. Show that these three functions span a space that L_3 and $L_\pm$ map into itself.

13.3 The multiplets on functions

Fix a length $s > 0$, and let

$$g(x) = e^{-|x|^2/2s^2} = g_1(x_1)\,g_1(x_2)\,g_1(x_3), \qquad g_1(y) = e^{-y^2/2s^2},$$

as in Proposition 8.3. We write $\mathcal{P}_n$ for the polynomials in x_1, x_2, x_3 that are homogeneous of degree n, and Q_N for those of degree at most N, so that $Q_N = \mathcal{P}_0 \oplus \cdots \oplus \mathcal{P}_N$. With the Hermite functions h_n of Lecture 8, for this s, let $h_{n_1n_2n_3}(x) = h_{n_1}(x_1)h_{n_2}(x_2)h_{n_3}(x_3)$, and let E_N be the span of the $h_{n_1n_2n_3}$ with $n_1 + n_2 + n_3 = N$. These spaces will turn out to be the levels of the oscillator in space (§ 13.6).

Theorem 13.5 (The multiplets on functions). *Let $n, N \geq 0$.*

1. *The space $\mathcal{P}_n g$ lies in $\mathcal{S}(\mathbb{R}^3)$, has dimension $(n+1)(n+2)/2$, and the L_j and every $U(R)$ map it into itself.*
2. *The $h_{n_1n_2n_3}$ form an orthonormal basis of $L^2(\mathbb{R}^3)$, and $Q_Ng = E_0 \oplus \cdots \oplus E_N$, an orthogonal sum. The L_j and every $U(R)$ map E_N into itself.*
3. *On E_N the L_j are an angular momentum, and E_N is an orthogonal sum of multiplets, exactly one of spin l for each $l = N, N-2, N-4, \cdots \geq 0$, and none of any other spin.*

So $L^2(\mathbb{R}^3)$ is an orthogonal sum of multiplets, and every spin that occurs is an integer.

Proof.

1. *Polynomials.* Every partial derivative of pg, for a polynomial p, is a polynomial times g, and a monomial times it is bounded. So $pg \in \mathcal{S}(\mathbb{R}^3)$. The monomials $x_1^{a_1}x_2^{a_2}x_3^{a_3}$ with $a_1+a_2+a_3 = n$ are a basis of $\mathcal{P}_n$, and there are $(n+1)(n+2)/2$ of them. By Proposition 13.4, $L_j(pg) = (L_j p)g$, and $L_1 p = -i\hbar(x_2\partial_3 p - x_3\partial_2 p)$ is again homogeneous of degree n, or zero. So are $L_2 p$ and $L_3 p$. Since $\left|R^{-1}x\right| = |x|$, $U(R)(pg) = (p\circ R^{-1})g$, and $p\circ R^{-1} \in \mathcal{P}_n$ because R^{-1} is linear.
2. *A basis.* By Theorem 8.4 and the standing fact S3, the $h_{n_1n_2n_3}$ form an orthonormal basis of $L^2(\mathbb{R}^3)$. By Proposition 8.3, $h_n = r_n g_1$ with r_n of degree n, so $h_{n_1n_2n_3} = r_{n_1}(x_1)r_{n_2}(x_2)r_{n_3}(x_3)\,g$, of degree $n_1+n_2+n_3$, with the leading term a positive multiple of $x_1^{n_1}x_2^{n_2}x_3^{n_3}$. By induction on the degree, every $x_1^{a_1}x_2^{a_2}x_3^{a_3}g$ with $a_1+a_2+a_3 \le N$ is a combination of the $h_{n_1n_2n_3}$ with $n_1+n_2+n_3 \le N$. So these span $Q_N g$, and $Q_N g = E_0 \oplus \cdots \oplus E_N$. By the first step, the L_j map $Q_N g$ and $Q_{N-1}g$ into themselves. For $\psi \in E_N$ and $\chi \in Q_{N-1}g$, Proposition 13.4 gives $\langle \chi, L_j\psi\rangle = \langle L_j\chi, \psi\rangle = 0$. So $L_j\psi$ is a vector of $Q_N g$ orthogonal to $Q_{N-1}g$, that is, a vector of E_N. The same holds for $U(R)$, with $\langle \chi, U(R)\psi\rangle = \langle U(R^{-1})\chi, \psi\rangle = 0$.
3. *An angular momentum.* In an orthonormal basis of E_N, the L_j are self-adjoint matrices, by the symmetry in Proposition 13.4, and they satisfy its commutation relations. By *Quantum Mechanics I*, Thm 7.3, E_N is an orthogonal sum of multiplets, and the number of spin l is $d_l - d_{l+1}$, where d_μ is the dimension of the eigenspace of L_3 in E_N for the eigenvalue $\hbar\mu$. (*Quantum Mechanics I* writes $\hbar m$; here m is the mass, so the label is μ.)
4. *Counting.* Since $x_1 = (w+\overline{w})/2$ and $x_2 = (w-\overline{w})/2i$, the monomials $w^k\overline{w}^{k'}x_3^{n-k-k'}$ with $k+k' \le n$ span $\mathcal{P}_n$, and there are $\dim \mathcal{P}_n$ of them, so they are a basis. By (13.1) and Proposition 13.4, L_3 is diagonal on $\mathcal{P}_n g$ in this basis, and its eigenvalue $\hbar\mu$ has the multiplicity

$$c_n(\mu) = \#\{(k,k') :\ k-k' = \mu,\ k+k' \le n\} = \begin{cases} \lfloor (n-|\mu|)/2\rfloor + 1, & \mu \in \mathbb{Z},\ |\mu| \le n, \\ 0, & \text{otherwise.}\end{cases}$$

   An eigenvector of L_3 in a direct sum of subspaces that L_3 maps into themselves has each of its components an eigenvector for the same eigenvalue, or zero. So in $Q_N g = \mathcal{P}_0 g \oplus \cdots \oplus \mathcal{P}_N g$ the eigenspace for $\hbar\mu$ has the dimension $\sum_{n\le N} c_n(\mu)$, and in $Q_N g = E_0 \oplus \cdots \oplus E_N$ it has the dimension $\sum_{n\le N} d_\mu(E_n)$. Subtracting the same identity for $N-1$ gives $d_\mu = c_N(\mu)$ in E_N. In particular $d_\mu = 0$ unless μ is an integer.
5. *The spins.* For an integer $l \ge 0$, $d_l - d_{l+1} = c_N(l) - c_N(l+1)$ is 1 if $l \le N$ and $N-l$ is even, and 0 otherwise. For a half-integer l, $d_l = d_{l+1} = 0$. With the second step, the multiplets of all the E_N together make up $L^2(\mathbb{R}^3)$. ∎

For example, E_1 is spanned by $x_1 g$, $x_2 g$ and $x_3 g$, one multiplet of spin 1. The orbital angular momentum of sodium's $3p$ electron (*Quantum Mechanics I*, §13.6) is a multiplet of spin 1 of this kind. The six functions of E_2 make one multiplet of spin 2 and one of spin 0. As a check on the count, $\sum_{l=N,N-2,\ldots}(2l+1) = (N+1)(N+2)/2 = \dim E_N$.

The standard bases of all the multiplets together form an orthonormal basis of $L^2(\mathbb{R}^3)$. We write $\mathcal{H}_l$ for the closed span of the multiplets of spin l, the vectors of *angular momentum* l.

Why integers? By (13.2), $U(R_3(\alpha))$ and the diagonal matrix $e^{-i\alpha L_3/\hbar}$ agree on a basis of $Q_N g$, so they agree on E_N. A rotation by 2π is the identity on functions, so $e^{-2\pi i\mu} = 1$ for every eigenvalue $\hbar\mu$ of L_3 on E_N, and μ is an integer. The same holds for every nonzero $\psi \in \mathcal{S}(\mathbb{R}^3)$ with $L_3\psi = \hbar\mu\psi$. Indeed, let e be a vector of the standard bases, with $L_3 e = \hbar\mu_e e$. Then the symmetry in Proposition 13.4 gives

$$\hbar\mu\,\langle e, \psi\rangle = \langle e, L_3\psi\rangle = \langle L_3 e, \psi\rangle = \hbar\mu_e\,\langle e, \psi\rangle\,,$$

so $\psi = 0$ unless μ is one of the integers μ_e. On a multiplet of half-integer spin, *Quantum Mechanics I*, §7.4, found $e^{-2\pi i J_z/\hbar} = -\mathbb{1}$ instead. So the spin $\frac{1}{2}$ of an electron is not an orbital angular momentum. It is not the $\mathbf{L}$ of any wave function on $\mathbb{R}^3$.

The relations alone do not give the theorem. The spin $\frac{1}{2}$ satisfies them too. The proof of *Quantum Mechanics I*, Thm 7.3, uses the finite dimension three times: to find a common eigenvector of J^2 and J_z, to stop the raising, since J_z has finitely many eigenvalues, and to stop splitting off multiplets. On $\mathcal{S}(\mathbb{R}^3)$, L_3 has infinitely many eigenvalues: by (13.1), $w^k g$ has $L_3 = \hbar k$ for every $k \geq 0$. What made the theorem apply is the finite-dimensional spaces E_N that the L_j keep.

As in Proposition 5.4, the standard bases and the numbers $\hbar\mu$ and $\hbar^2 l(l+1)$ make two self-adjoint operators, which we still call L_3 and L^2. Let e_k run through these basis vectors, with labels l_k and μ_k, and put $c_k = \langle e_k, \psi\rangle$. Their domains are

$$D(L_3) = \Big\{\psi : \sum_k \mu_k^2 |c_k|^2 < \infty\Big\}, \qquad D(L^2) = \Big\{\psi : \sum_k l_k^2 (l_k+1)^2 |c_k|^2 < \infty\Big\}.$$

These operators extend the earlier rules on $\mathcal{S}(\mathbb{R}^3)$. As in the proof of Theorem 8.8, they act on $\mathcal{S}(\mathbb{R}^3)$ by the rules of Definition 13.3. By Postulate 5.5, a reading of L_3 gives $\hbar\mu$ with μ an integer, and a reading of L^2 gives $\hbar^2 l(l+1)$ with l an integer.

Self-check 13.3

1. Use the counting step of the proof to show that $\mathcal{P}_2 g$ holds one multiplet of spin 2 and one of spin 0, and find the vector of spin 0.

13.4 Spherical harmonics

The multiplets of spin l can be represented by polynomials. The proof of Theorem 13.5 applies to $\mathcal{P}_l g$ itself, a finite-dimensional space that the L_j keep, with $d_\mu = c_l(\mu)$. So $\mathcal{P}_l g$ holds one multiplet of each spin $l, l-2, \ldots$. Let $\mathcal{V}_l \subseteq \mathcal{P}_l$ be the polynomials whose products with g make up the multiplet of spin l. In $\mathcal{P}_l$, the eigenvalue $\hbar l$ of L_3 belongs only to w^l, since $k - k' = l$ and $k + k' \leq l$ force $k = l$. So w^l is the top vector of $\mathcal{V}_l$, and $\mathcal{V}_l$ is spanned by $L_-^i w^l$, $i = 0, \ldots, 2l$. The radial part of the Laplacian separates from L^2 as follows. Let $D = x_1\partial_1 + x_2\partial_2 + x_3\partial_3$.

Lemma 13.6 (The Laplacian and L^2). *For every twice continuously differentiable function f on an open set of $\mathbb{R}^3$,*

$$r^2 \Delta f = D^2 f + Df - \frac{1}{\hbar^2} L^2 f.$$

Proof. By Definition 13.3, $-L^2/\hbar^2$ is the sum of the squares of $x_j\partial_k - x_k\partial_j$ over the pairs $j < k$. For one pair,

$$(x_1\partial_2 - x_2\partial_1)^2 f = x_1^2\partial_2^2 f + x_2^2\partial_1^2 f - 2x_1x_2\partial_1\partial_2 f - x_1\partial_1 f - x_2\partial_2 f.$$

Summed over the three pairs, this is $\sum_{j\neq k} x_j^2\partial_k^2 f - 2\sum_{j<k} x_j x_k \partial_j\partial_k f - 2Df$. On the other side, $r^2\Delta f = \sum_{j,k} x_j^2\partial_k^2 f$ and $D^2 f = \sum_{j,k} x_j x_k \partial_j\partial_k f + Df$. In $r^2\Delta f - D^2 f - Df$ the terms with $j = k$ cancel, and what remains is the sum above. ∎

For $p \in \mathcal{P}_l$, $Dp = lp$, so the lemma gives $r^2\Delta p = \big(l(l+1) - L^2/\hbar^2\big)p$. The other multiplets of $\mathcal{P}_l g$ have spins below l, so p is *harmonic*, $\Delta p = 0$, exactly when $L^2 p = \hbar^2 l(l+1)p$, that is,

when $p \in \mathcal{V}_l$. So $\mathcal{V}_l$ is the space of harmonic polynomials homogeneous of degree l, the space Hall, §17.6, starts from.

Figure 13.1 separates the radius from the direction.

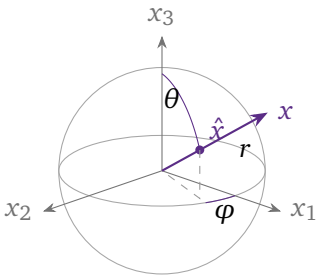


Figure 13.1: A point $x = r\hat{x}$ is specified by its radius r and its direction $\hat{x}$ on the unit sphere. The angles θ and φ describe the direction; the spherical harmonics depend on these angles.

For $p \in \mathcal{V}_l$, the function $p(x)/|x|^l$ depends only on the direction of x. In spherical angles, $x = r(\sin\theta\cos\varphi,\ \sin\theta\sin\varphi,\ \cos\theta)$, the top vector gives $w^l/r^l = \sin^l\theta\, e^{il\varphi}$. Normalised on the unit sphere S^2, with the inner product $\langle Y, Z\rangle = \int_0^{2\pi}\int_0^{\pi} \overline{Y}Z \sin\theta\,\mathrm{d}\theta\,\mathrm{d}\varphi$, and lowered with the Condon–Shortley coefficients of *Quantum Mechanics I*, Thm 7.3, these functions are the *spherical harmonics*. The constant comes from $\int_0^{\pi}\sin^{2l+1}\theta\,\mathrm{d}\theta = 2^{2l+1}(l!)^2/(2l+1)!$, found by integration by parts:

$$Y_{ll} = (-1)^l \frac{\sqrt{(2l+1)!/4\pi}}{2^l\, l!}\, \sin^l\theta\, e^{il\varphi}, \qquad L_- Y_{l\mu} = \hbar\sqrt{(l+\mu)(l-\mu+1)}\, Y_{l,\mu-1}.$$

The sign $(-1)^l$ is the usual choice of phase. For example, $Y_{00} = 1/\sqrt{4\pi}$,

$$Y_{10} = \sqrt{\frac{3}{4\pi}}\cos\theta, \qquad Y_{1,\pm1} = \mp\sqrt{\frac{3}{8\pi}}\sin\theta\, e^{\pm i\varphi},$$

and rY_{10}, $rY_{1,\pm1}$ are multiples of x_3 and of $x_1 \pm ix_2$. Teschl, (10.40), gives every $Y_{l\mu}$ by the associated Legendre functions.

Theorem 13.7 (Spherical harmonics · *cited*)**.** *The $Y_{l\mu}$, $l \geq 0$, $|\mu| \leq l$, form an orthonormal basis of $L^2(S^2)$. Every $\psi \in L^2(\mathbb{R}^3)$ is a sum, convergent in norm,*

$$\psi(x) = \sum_{l,\mu} \frac{u_{l\mu}(r)}{r}\, Y_{l\mu}(\theta,\varphi), \qquad \|\psi\|^2 = \sum_{l,\mu}\int_0^{\infty} \left|u_{l\mu}(r)\right|^2 \mathrm{d}r,$$

and $\psi \mapsto (u_{l\mu})$ is a unitary map onto the sequences of functions in $L^2(0,\infty)$ with a finite sum of squared norms.

This is Hall, Thm 17.12 and p. 382, and Teschl, Thm 10.7 and (10.45)–(10.49). Both proofs approximate continuous functions by polynomials (Hall by the Stone–Weierstrass theorem, Teschl by Weierstrass's), and the decomposition of $L^2(\mathbb{R}^3)$ integrates square-integrable functions in spherical coordinates. This course does neither. Theorem 13.5 gives the decomposition of $L^2(\mathbb{R}^3)$ into multiplets without them.

Self-check 13.4

1. Show from Lemma 13.6 that $|x|^2$ is not harmonic, and compute $L^2\,|x|^2$ and $\Delta\,|x|^2$.

The physics

A molecule of HCl rotates almost as a rigid body: the distance d between the two nuclei hardly changes. The direction of its axis is a point of S^2, and its energy of rotation is $L^2/2I$, with the moment of inertia $I = md^2$ and the reduced mass m of the two nuclei. The $Y_{l\mu}$ have $L^2 = \hbar^2 l(l+1)$ and, by Theorem 13.7, form a basis of $L^2(S^2)$, so the levels are $\hbar^2 l(l+1)/2I$, each $2l+1$ times. Absorbing a photon takes the molecule from l to $l+1$ (a rule of the coupling to light, not derived here), with the energy $\hbar^2(l+1)/I$. So the lines are equally spaced, at $l+1$ times the frequency $\hbar/2\pi I$. The lines of $H^{35}Cl$ lie at 625.9, 1251.5, 1876.2 and 2499.9 GHz (Nolt and coworkers, 1987), within 0.2% of the multiples of 625.9 GHz. They fall slightly short because the spinning molecule stretches. From $\hbar/2\pi I = 625.9\,\text{GHz}$, $I = 2.68 \times 10^{-47}\,\text{kg m}^2$, and with $m = 0.980\,\text{u}$, $d = 1.28\,\text{Å}$, an average over the molecule's vibration (the equilibrium distance is 1.27 Å).

13.5 Central potentials

A potential is *central* when it depends only on $r = |x|$: the free particle ($V = 0$), the oscillator ($V = m\nu^2 r^2/2$) and hydrogen ($V = -C/r$, Lecture 14). The rule of $H_0 + V$ on smooth functions is $-(\hbar^2/2m)\Delta + V(|x|)$; which smooth functions lie in its domain depends on V.

Proposition 13.8 (The radial rule). *Let V be a real continuous function on $(0,\infty)$, $l \geq 0$ an integer, $p \in \mathcal{V}_l$, and u a smooth function on $(0,\infty)$. Then $\psi(x) = u(|x|)\,p(x)/|x|^{l+1}$ satisfies, for every $x \neq 0$,*

$$-\frac{\hbar^2}{2m}\Delta\psi + V(|x|)\psi = (H_l u)(|x|)\,\frac{p(x)}{|x|^{l+1}}, \qquad H_l u = -\frac{\hbar^2}{2m}u'' + \Big(\frac{\hbar^2 l(l+1)}{2mr^2} + V(r)\Big)u.$$

If u vanishes outside a closed bounded interval $[a,b] \subset (0,\infty)$, then ψ, with $\psi(0) = 0$, is smooth on $\mathbb{R}^3$ and vanishes for $|x| < a$ and $|x| > b$.

Proof. On $|x| > 0$, write $\psi = fY$ with $f = u(r)/r$ and $Y = p/r^l$. Since $Y(tx) = Y(x)$ for $t > 0$, differentiating in t at $t = 1$ gives $DY = 0$. Since $\partial_j r = x_j/r$, $Df = rf'$. A smooth function $F(r)$ on $|x| > 0$ is $G(|x|^2)$ with $G(t) = F(\sqrt{t})$ smooth for $t > 0$, so by Proposition 13.4 the L_j commute with multiplication by it, and $L^2\psi = fr^{-l}L^2 p = \hbar^2 l(l+1)\psi$. By the product rule, $D\psi = rf'Y$ and $D^2\psi = r(rf')'Y$. So Lemma 13.6 gives

$$r^2\Delta\psi = \big(r^2 f'' + 2rf' - l(l+1)f\big)Y.$$

With $f = u/r$, $f'' + 2f'/r = u''/r$, so $\Delta\psi = (u'' - l(l+1)u/r^2)\,Y/r$. Multiplying by $-\hbar^2/2m$ and adding $V\psi$ gives the identity. Away from 0, ψ is smooth as a composition of smooth functions, and in the second case it vanishes near 0. ∎

The rule H_l is the one-dimensional Hamiltonian of Lecture 10 on the half-line, with the extra *centrifugal term* $\hbar^2 l(l+1)/2mr^2$. For $l \geq 1$ it is a repulsion near the origin: a particle with angular momentum keeps away from the centre. By Theorem 13.7, on the vectors $(u(r)/r)Y_{l\mu}$ with l and μ fixed, the map to u is unitary onto $L^2(0,\infty)$. Lecture 15 states, for the free particle and for hydrogen, which self-adjoint operator on $L^2(0,\infty)$ the Hamiltonian becomes on these vectors, and the condition $u(0) = 0$ for $l = 0$ that goes with it.

Self-check 13.5

1. Let $u(r) = \sin kr$ with $k > 0$. Show that $H_l u = (\hbar^2 k^2/2m)u$ for $l = 0$ and $V = 0$, and that $\sin(k|x|)/|x|$ is not square integrable.

13.6 The oscillator in space

Definition 13.9 (The oscillator in space)**.** The Hamiltonian H of the isotropic oscillator of mass m and frequency ν, with $s = \sqrt{\hbar/m\nu}$, is the operator made, as in Proposition 5.4, from the basis $(h_{n_1 n_2 n_3})$ and the numbers $\hbar\nu(n_1 + n_2 + n_3 + \frac{3}{2})$.

By Proposition 5.4 it is self-adjoint. The rule $-(\hbar^2/2m)\Delta + m\nu^2|x|^2/2$ is the sum of three rules (8.1), one in each variable, so it maps $h_{n_1 n_2 n_3}$ to $\hbar\nu(n_1 + n_2 + n_3 + \frac{3}{2})h_{n_1 n_2 n_3}$, by Proposition 8.3. As in the proof of Theorem 8.8, H acts by this rule on $\mathcal{S}(\mathbb{R}^3)$. The eigenspace of H for $\hbar\nu(N + \frac{3}{2})$ is E_N, since a vector with this eigenvalue has no component on the other basis vectors.

Example 13.10 (The level N counted twice)**.**

1. *By coordinates.* The level $\hbar\nu(N + \frac{3}{2})$ has one basis vector for each triple (n_1, n_2, n_3) with $n_1 + n_2 + n_3 = N$: for each n_1 there are $N - n_1 + 1$ choices of n_2, so $(N + 1)(N + 2)/2$ in all.
2. *By rotations.* By Theorem 13.5, E_N holds one multiplet of each spin $l = N, N - 2, \ldots$, so its dimension is $\sum(2l + 1)$ over these l, an arithmetic progression with the step 4, whose sum is again $(N + 1)(N + 2)/2$. For example, the level $N = 2$ has $6 = 5 + 1$ states, of spins 2 and 0.

Each level E_N is a sum of whole multiplets (Theorem 13.5), as *Quantum Mechanics I*, Cor. 7.5, leads one to expect of a Hamiltonian that commutes with rotations. That the spins N and $N - 2$ share a level is not forced by rotations. It comes from a larger symmetry of the oscillator, not used here, and Lecture 15 meets the same in hydrogen.

Self-check 13.6

1. How many states of the oscillator have energy at most $\hbar\nu(N + \frac{3}{2})$? Which spins occur at the level $N = 4$?

A particle in space

States: $L^2(\mathbb{R}^3)$ is an orthogonal sum of finite-dimensional multiplets of $\mathbf{L}$, made from polynomials times a Gaussian, and every spin is an integer (Theorem 13.5). *Observables:* the components of $\mathbf{L} = Q \times P$ satisfy $[L_1, L_2] = i\hbar L_3$ on $\mathcal{S}(\mathbb{R}^3)$ (Proposition 13.4). Away from the origin, the rule of a central Hamiltonian on $(u(r)/r)Y_{l\mu}$, with u smooth, is the radial rule H_l with the centrifugal term (Proposition 13.8). *Measurement:* a reading of L_3 gives $\hbar\mu$ with μ an integer, and a reading of L^2 gives $\hbar^2 l(l + 1)$ with l an integer; never $\hbar/2$.

Summary.

- *Three dimensions*: Q_j, P_j and H_0 are self-adjoint and act on $\mathcal{S}(\mathbb{R}^3)$ by x_j, $-\mathrm{i}\hbar\partial_j$ and $-(\hbar^2/2m)\Delta$ (Proposition 13.2).
- *Orbital angular momentum*: $\mathbf{L} = Q \times P$ is symmetric on $\mathcal{S}(\mathbb{R}^3)$ and satisfies $[L_1, L_2] = \mathrm{i}\hbar L_3$ (cyclic), and L_3 generates the rotations about the x_3-axis (Proposition 13.4).
- *The multiplets on functions*: the oscillator's level E_N is a sum of one multiplet of each spin $N, N-2, \ldots$, and these make up $L^2(\mathbb{R}^3)$; only integer spins occur (Theorem 13.5).
- *Spherical harmonics*: the multiplet of spin l in $\mathcal{P}_l$ is the harmonic polynomials, with top vector $(x_1 + \mathrm{i}x_2)^l$ (Lemma 13.6); the $Y_{l\mu}$ form a basis of $L^2(S^2)$ (Theorem 13.7, cited).
- *Central potentials*: away from the origin, the rule of $H_0 + V$ on $(u(r)/r)Y_{l\mu}$, with u smooth, is the radial rule with the term $\hbar^2 l(l+1)/2mr^2$ (Proposition 13.8).

Lecture 15 solves the radial rule for hydrogen.

Self-check answers.

13.1 1. By the chain rule, $\partial_j(\psi \circ R^{-1}) = \sum_k (R^{-1})_{kj}(\partial_k\psi) \circ R^{-1}$. Applying it twice and summing over j gives $\sum_j (R^{-1})_{kj}(R^{-1})_{k'j} = \delta_{kk'}$, since R^{-1} is a rotation. So $\Delta(\psi \circ R^{-1}) = (\Delta\psi) \circ R^{-1}$, and $H_0 = -(\hbar^2/2m)\Delta$ on $\mathcal{S}(\mathbb{R}^3)$ by Proposition 13.2.

13.2 1. $L_3 w = \hbar w$, $L_3\overline{w} = -\hbar\overline{w}$, $L_3 x_3 = 0$; $L_+ w = 0$, $L_+ x_3 = -\hbar w$, $L_+\overline{w} = 2\hbar x_3$; and $L_- w = -2\hbar x_3$, $L_- x_3 = \hbar\overline{w}$, $L_-\overline{w} = 0$. Since $L_1 = (L_+ + L_-)/2$ and $L_2 = (L_+ - L_-)/2\mathrm{i}$, every L_j maps the span of x_1, x_2, x_3 into itself: a multiplet of spin 1.

13.3 1. Here $c_2(0) = 2$ and $c_2(\pm 1) = c_2(\pm 2) = 1$, so $d_0 - d_1 = 1$, $d_1 - d_2 = 0$ and $d_2 - d_3 = 1$. The vector of spin 0 is $|x|^2 g$, which every L_j maps to 0 by Proposition 13.4.

13.4 1. Here $D|x|^2 = 2|x|^2$ and, by Proposition 13.4, $L^2|x|^2 = 0$. So $r^2\Delta|x|^2 = (4+2)|x|^2$, and $\Delta|x|^2 = 6 \neq 0$: $|x|^2$ has spin 0, not 2.

13.5 1. $-(\hbar^2/2m)(\sin kr)'' = (\hbar^2 k^2/2m)\sin kr$. The squared norm is $\int_0^\infty 4\pi r^2(\sin^2 kr/r^2)\,\mathrm{d}r = 4\pi\int_0^\infty \sin^2 kr\,\mathrm{d}r$, which is infinite: like the plane waves, the spherical wave is not a state.

13.6 1. $\sum_{n\le N}(n+1)(n+2)/2 = (N+1)(N+2)(N+3)/6$ states. At $N = 4$ the spins are 4, 2 and 0, with $9 + 5 + 1 = 15$ states.

References

Hall, *Quantum Theory for Mathematicians*, §17.6 (pp. 376–380), Thm 17.12 (p. 377), Prop. 17.19 (pp. 381–382). Teschl, *Mathematical Methods in Quantum Mechanics*, (10.40) and Thm 10.7 (p. 227), (10.45)–(10.49) (p. 229). I. G. Nolt et al., *J. Mol. Spectrosc.* 125, 274 (1987), as listed in the JPL Molecular Spectroscopy catalogue, entry 36001.